SCIENCE

OBJECTIVES:

The objective of teaching science to Class X students is to:

1. Provide, through well designed studies of the experimental and practical science, a worthwhile educational experience for all students, whether or not they intend to study science beyond the secondary stage. In particular, it aims at enabling them to acquire sufficient understanding and knowledge to:

1.1 Become confident citizens in a technological world, and enable them to take active interest in matters of scientific importance

1.2 Recognize the usefulness and limitations of scientific method and to appreciate its applicability in other disciplines and in everyday life.

1.3 Be suitably prepared for studies beyond the secondary stage in pure sciences or in applied sciences or in science – dependent courses.

2. Develop abilities and skills that:

2.1 are relevant to the study and practice of science.
2.2 are useful in everyday life.
2.3 encourage efficient and safe practice.
2.4 encourage effective communication.

3. Develop Attitude relevant to science such as:

3.1 concern for accuracy and precision.
3.2 objectivity
3.3 integrity
3.4 enquiry
3.5 initiatives
3.6 inventiveness/ innovation.

4. Stimulate interest in and care for the environment

5. Promote awareness that:

5.1 Scientific theories and methods have developed and continue to do so as a result of cooperative activities of groups and individuals.

5.2 The study and practice of science are subject to social, economic, technological ethical and cultural influences and limitations.

5.3 The application of science may be both beneficial and detrimental to the individual, the community and the environment.

5.4. Science transcends national boundaries and that the language of science, correctly and rigorously applied is universal

Domains

The three domains in Science (Physics, Chemistry and Biology) are:

(a) Knowledge with understanding.
(b) Handling information and solving problems.
(c) Experimental skills and investigations.

Description of each domain is given below.

(a) Knowledge with understanding

1. Scientific phenomena, facts, laws, definitions, concepts, theories.
2. Scientific vocabulary, terminology, conventions (including symbols, quantities and units).
3. Scientific instruments and apparatus including techniques of operation and aspects of safety.
4. Scientific quantities and their determination.
5. Scientific and technological applications with their social, economic and environmental implications.

The curriculum objectives define the factual material that candidates may be required to recall and explain. Question testing these objectives will often begin with one of the following words; define state, describe, explain or outline etc.

(b) Handling Information and Solving Problems

Students should be able, in words or using other written forms of presentation (i.e. symbolic, graphical and numerical) to:

1. locate, select, organize and present information from a variety of sources;
2. translate information from one to another;
3. manipulate numerical and other data;
4. use information to identify patterns, report trends and draw inferences;
5. present reasoned explanations for – phenomena, patterns and relationships;
6. make predictions and hypotheses;
7. solve problems;

These skills cannot be precisely specified in the curriculum objectives because the questions testing such skills are often based on information which is unfamiliar to the candidate. In answering such questions, candidates are required to use principles and concepts that are within the syllabus and apply them in a logical, deductive manner to a novel situation.

The questions for testing these skills, will often begin with one of the following words; Discuss, predict,suggest, calculate or determine etc.

(c) Experimental Skills and Investigations

Students should be able to

1. use techniques, apparatus and materials;
2. make and record observations, measurements and estimates;
3. interpret and evaluate experimental observations and date;
4. plan, investigate and/or evaluate methods and suggest possible improvements (including the selection of techniques, apparatus and materials).

Scheme of Assessment

Board Examination

Theory

Theory paper of Science shall have 80 marks spread over one single question paper containing three sections, viz, Section – I based on the syllabus of Physics and Section – II based on the syllabus of Chemistry and Section – III based on the syllabus of Biology having a weightage of 80 marks and 3 hours duration.

Section I (Physics)	:	**26 marks**
Section II (Chemistry)	:	**26 marks**
Section III (Biology)	:	**28 marks**

Each section shall contain

(i)	03 long answer questions with internal choice each of 05 marks	**3 x 5 = 15 marks**
(ii)	09 short answer questions with internal choice each of 03 marks	**9 x 3 = 27 marks**
(iii)	10 very short answer question each of 02 marks	**10 x 2 = 20 marks**
(iv)	18 multiple choice questions each of 01 mark	**18 x 1 = 18 marks**

Total marks for Physics, Chemistry and Biology Sections: 80 marks

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Theory Marks: 80 (Physics + Chemistry + Biology: 26+26+28) **Time allotted: 3 hrs**

Internal Assessment: 20 Marks

Section – I (Physics)

Unit 1:- Light- Reflection and Refraction **Marks 08**

Reflection of light, spherical mirrors; image formation; use of spherical mirrors.

Sign conventions for spherical mirrors; relation between focal length and radius of curvature; mirror formula (only relation) (Numerical Problems); magnification.

Refraction of light, refraction through a glass slab, refractive index, Conditions for no refraction.

Spherical lenses, image formation, sign conventions, lens formula (only relation) (Numerical Problems), Magnification (Numerical Problems), Power of a lens (Numerical Problems).

Unit 2: The Human Eye and the Colorful World **Marks 05**

Human eye; power of accommodation; defects of vision and their correction.

Glass prism (refraction and dispersion).

Atmospheric refraction – twinkling of stars and color of sun at sunrise and sunset.

Unit 3 :-Electricity **Marks 06**

Concept of electric charge;

Electric current; electric potential and potential difference;

Ohm's law and experimental verification; resistance and its dependence; combination of resistances (series and parallel) (Numerical Problems)

Heating effect of current – Electric power and energy (Numerical Problems)

Unit 4: Magnetic Effects of Current **Marks 07**

Oersted's Experiment; Magnetic field and field lines.

Magnetic field due to a current carrying conductor- Straight, coil (loop) and solenoid (qualitativeonly).

Force on a current carrying conductor in a magnetic field.

Domestic electric circuits.

SECTION-II (CHEMISTRY)

Unit-I **Chemical Reactions and Equation** **Marks 06**

Chemical equation, writing of chemical equation; Balancing chemical equations.

Types of chemical reactions, viz. Combination reactions; Decomposition reactions; Displacement reactions; Double displacement reactions; Oxidation and reduction.

Effects of oxidation and reduction reactions in everyday life, viz, corrosion and rancidity.

Unit-II **Carbon and its Compounds** **Marks 08**

Bonding in Carbon, Covalent bond, Allotropes of carbon;

Versatile nature of carbon; Saturated and unsaturated hydrocarbons; chains; branches and rings, homologous series and its characteristics; nomenclature of Carbon compounds.

Chemical properties of carbon compounds viz. combustion; oxidation; addition and substitution reactions.

Important Carbon compounds like Ethanol and Ethanoic acid. Properties of Ethanol and Ethanoic acid.

Soaps and Detergents.

Unit-III **Metals and Non-metals** **Marks 07**

Physical properties of metals and non- metals.

Chemical properties of metals like action of water, air, acids, salts; Reactivity series of metals.

Cause of reactivity of metals and non- metals. Properties of Ionic compounds.

Occurrence of metals; their extraction, enrichment of ores. Extraction of metals in accordance with activity series; refining of metals.

Corrosion of metals and its prevention.

Unit-IV **Acids, Bases and Salts** **Marks 05**

Idea about acids and bases; chemical properties of acids and bases viz. Action of metals, metal carbonates, metal hydrogen carbonates (only in case of acids), metallic acids; non- metallic acids and bases.

Similarities in acids and bases; reaction of acids and bases with water.

Strength of acid and base solutions; pH, its importance.

Idea of salts, their family and pH.Chemicals from common salts like Sodium hydroxide, Baking soda and Washing soda; Hydrated salts Plaster of Paris.

SECTION – III (BIOLOGY)

Unit 1: - Life Processes **Marks 08**

- What are life Processes?
- Nutrition – Autotrophic Nutrition, Heterotrophic Nutrition. How do animals obtain their nutrition? Nutrition in Human beings.
- Respiration.
- Transportation: Transportation in Plants.
- Excretion: Excretion in Human beings, Excretion in Plants.

Unit 2:- Control and Co-ordination **Marks 06**

- Animals – Nervous System, What happens in Reflex Action? Human Brain; How are these tissues protected? How does Nervous tissue cause action?
- Coordination in Plants, immediate response to stimulus, movements due to growth.
- Hormones in Animals.

Unit 3 :- How do Organism Reproduce **Marks 06**

- Do organisms create exact copies of themselves? The importance of variation.
- Modes of reproduction used by unicellular organisms. Fission. Fragmentation, Regeneration. Vegetative Propagation, Budding, Spore Formation.
- Sexual Reproduction: Why the sexual mode of reproduction? Sexual reproduction in flowering plants, Reproduction in human Beings. Male Reproductive System, Female Reproductive System. What happens when the egg is not fertilized? Reproductive Health.

Unit 4: Heredity **Marks 03**

- Accumulation of variation during reproduction.
- Heredity: Inherited traits, Rules for the inheritance of traits, Mendel's contribution. How do thesetraits get expressed? Sex determination.

Unit 5: Our Environment **Marks 05**

- What happens when we add our waste to the Environment?
- Ecosystem – What are its components? Food chains and Webs.
- How do our activities effect the environment? Ozone layer and how it is getting depleted. Managing the garbage we produce.

PRACTICAL WORK

Section – I

(PHYSICS)

List of Practicals

Section 1

1. Verify the laws of reflection using mirror strips.
2. Verify the laws of refraction using glass slab, and find the refractive index of the material.
3. Show that the image is as far behind the plane mirror as the object is in front of it.
4. Trace the path of rays through a glass prism and measure the angle of deviation. (2 obs).
5. Using a glass prism, Prove that $\angle i+\angle e=\angle A+\angle d$
6. Find the focal length of convex mirror using the candle wire gauge and screen.
7. Trace the path of a ray of light through a glass slab and show that the angle of incidence is equal to the angle of emergence.
8. Verify Ohm's law (3 observations).
9. Prepare an electromagnet and observe the relation between the direction of current and polarity of the electromagnet.
10. Draw the magnetic lines for a bar magnet using a compass needle.

Project work

1. To construct an Astronomical Telescope using two convex lenses.
2. Prepare a model showing the production of thermo electric energy.
3. To generate wind energy using a wind mill.
4. Prepare an electromagnet.
5. To prepare a model to demonstrate Oersted's experiment.

SECTION – II

(CHEMISTRY)

1. Determine the pH of water obtained from different sources (at least four)
2. Determine the pH of juice obtained from different fruits.
3. Simple volumetric acid – base titrations.
4. Preparation of soap.
5. Preparations of hydrogen gas by the action of diluted sulphuric acid oil and study the properties.
6. To test the conductivity of various salt solutions.

Project work

1. Make a list that contains.
   Five elements in the free state and
   Five elements in the combined state.
2. To frame modern periodic table.
3. To make models of alkanes.

SECTION III
(BIOLOGY)

1. Identification of organs through charts/ Models of the following.
    a. Digestive system in humans.
    b. Circulatory system in humans.
    c. Nervous system in human beings.
    d. Respiratory system in human beings.
2. To study the reproductive organs of a flower.
3. To measure the body temperature and observe pulse rate.
4. To measure the blood pressure using sphagnometer.
5. To show that green plants synthesize carbohydrates.
6. To prepare and study the temporary mount of Lactobacillus.
7. To study the microorganism in stagnant water under microscope.
8. To study Mendelian ratios using pea seeds.

Project work

a. Preparation of vermin-compost (earth worms).
b. Making a herbarium (At least 10 aquatic and 10 terrestrial plants).
c. Effects of water pollution

Or

Air Pollution

Or

Noise pollution

Prescribed Textbook:

A Textbook of Science for Class X Published by J&K Board of School Education.